

Distribuciones Discretas en Julia

Nombre	Parámetros	$P(X=k)=f(k)$	Media $E(X)$	Varianza $V(X)$	fgm $E(e^{tX})$
Categorical $C(p)$	vector $p, \sum p_i = 1$ $x = 1, 2, 3, \dots, k$	$P(X = k) = p[k]$	$\sum_{i=1}^k i p_i$	$\sum_{i=1}^k i^2 p_i - \left(\sum_{i=1}^k i p_i\right)^2$	
Discrete NonParametric		soporte vector x probabilidades p $P(X = x_i) = p_i$	$\sum x_i p_i$	$\sum x_i^2 p_i - \left(\sum x_i p_i\right)^2$	
Binomial $b(n, p)$	$n \in \mathbb{N}, p \in (0, 1)$ $p + q = 1$	$\binom{n}{k} p^k q^{n-k}$ $k \in \{0, \dots, n\}$	np	npq	$(q + pe^t)^n$
Geométrica $G(p)$	$p \in (0, 1)$ $p + q = 1$	pq^k $k \in \{0, 1, 2, \dots\}$	$\frac{q}{p}$	$\frac{q}{p^2}$	$\frac{p}{1 - qe^t}$ $t < -\ln q$

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Poisson $p(\lambda)$	$\lambda > 0$	$\frac{\lambda^k e^{-\lambda}}{k!}$ $k \in \{0, 1, 2, \dots\}$	λ	λ	$e^{\lambda(e^t-1)}$
Hipergeo- métrica $H(s, f, n)$	$n \in \mathbb{N}$ $s \in \mathbb{N}$ $f \in \mathbb{N}$ $n < s + f$	$\frac{\binom{s}{x} \binom{f}{n-x}}{\binom{s+f}{n}}$ $x \in \{\max(0, n - f), \dots, \min(n, s)\}$	$n \frac{s}{s+f}$	$n \left(\frac{s}{s+f} \right) \left(\frac{f}{s+f} \right) \left(\frac{s+f-n}{s+f-1} \right)$	
Binomial Negativa $bn(p, r)$	$r \in \mathbb{N}$ $p \in (0, 1)$	$\binom{k+r-1}{k} p^r q^k$ $x \in \{0, 1, 2, \dots\}$	$r \frac{q}{p}$	$\frac{rq}{p^2}$	$\left[\frac{p}{1 - qe^t} \right]^r$ $t < -\ln q$